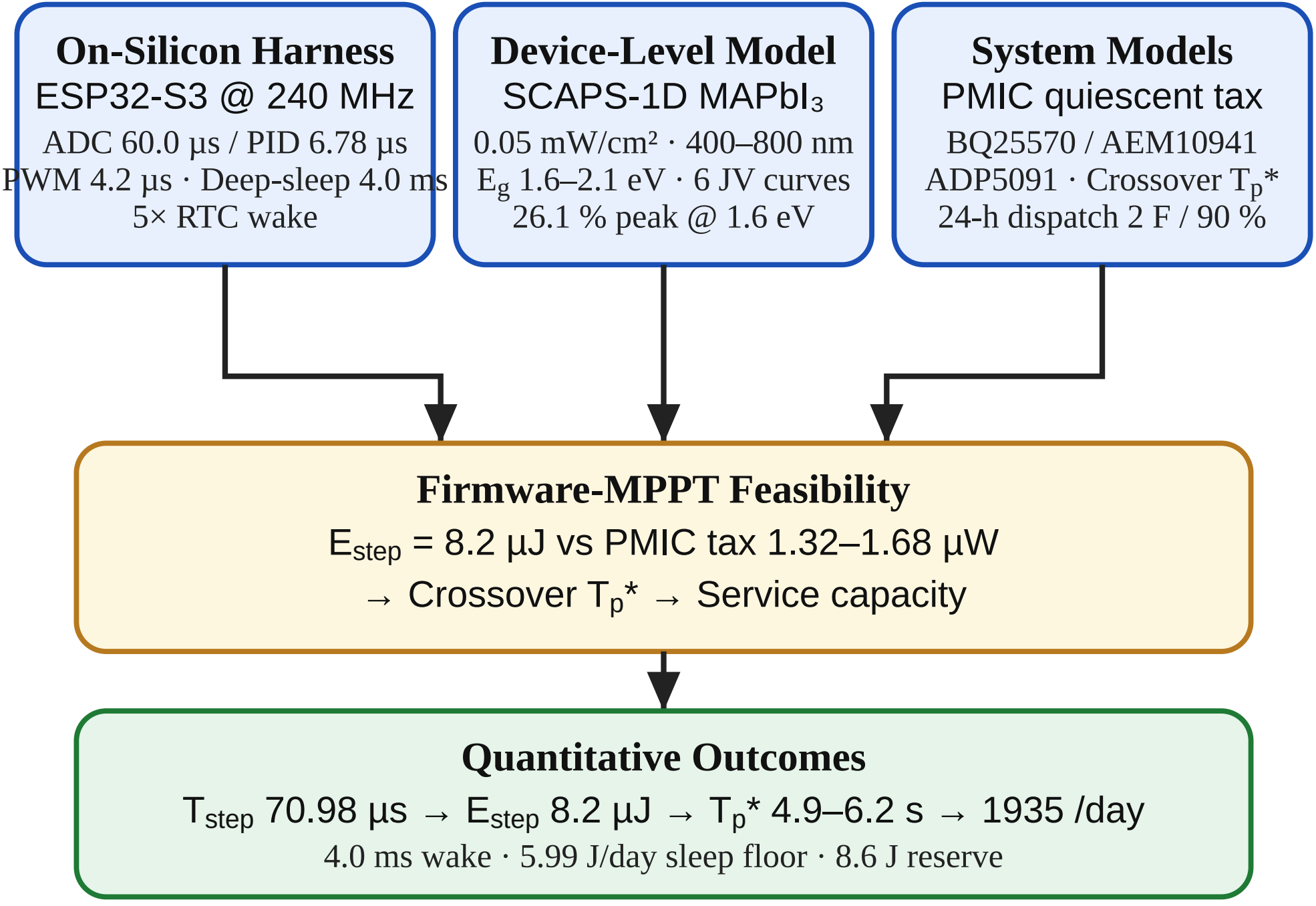
On-Silicon Harness
ESP32-S3 @ 240 MHz
ADC 60.0 µs / PID 6.78 µs
PWM 4.2 µs · Deep-sleep 4.0 ms
5× RTC wake
Device-Level Model
SCAPS-1D MAPbI3
0.05 mW/cm² · 400–800 nm
Eg 1.6–2.1 eV · 6 JV curves
26.1 % peak @ 1.6 eV
System Models
PMIC quiescent tax
BQ25570 / AEM10941
ADP5091 · Crossover Tp*
24-h dispatch 2 F / 90 %
Firmware-MPPT Feasibility
Estep = 8.2 µJ vs PMIC tax 1.32–1.68 µW
→ Crossover Tp* → Service capacity
Quantitative Outcomes
Tstep 70.98 µs → Estep 8.2 µJ → Tp* 4.9–6.2 s → 1935 /day
4.0 ms wake · 5.99 J/day sleep floor · 8.6 J reserve